

Who Let The Bot In?

Control Agentic AI Before It Goes Too Far.

Hi, what do you want to get done today?



Ok I need to create a plan to manage all the AI agents being used across my organization. I'm not quite sure where to even start. I have some employees asking me to create service accounts for their agents to use, while others seem to be just sharing their own credentials directly. Provide for me an overview of how AI agents work, what best practices there are to manage them, and how to make sure employees are being safe with them.



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I've been hearing a lot about agentic AI... is it time to do something about it?



Agentic AI isn't just another step forward in artificial intelligence, it's a different kind of player entirely. **These systems are autonomous.** That means they operate on their own, set and pursue goals, adapt as they learn, and can take action *without asking for permission*. That shift from following instructions to acting with autonomy changes the rules for how we manage and secure technology.

For years, we've built our security and identity models around a simple assumption: software does what we tell it to. But agentic AI breaks that pattern. These systems don't just follow commands. They behave more like digital coworkers, capable of making decisions and learning from outcomes in real time.

This level of autonomy brings powerful benefits. But it also introduces a new kind of risk — one that our current systems aren't built to manage.

Right now, agentic AI adoption is accelerating faster than our ability to govern it:

- While 82% of organizations are already using AI agents and 96% of IT leaders recognize them as a growing security threat...
- Only 44% have any formal policies in place to manage how these agents operate.^{1,2}

That disconnect between adoption and governance is what this eBook tackles head-on. Even if AI behaves like a human, it cannot reason like one, which introduces risks that demand oversight.

To manage agentic AI safely, *we need a mindset shift*. We must treat autonomous AI agents like digital identities, with the same (or even tighter) oversight and safeguards we'd apply to any employee or contractor.

That includes assigning them a unique identity, limiting their access from day one, monitoring their behavior, and applying Zero Trust principles at every step.

If your organization is already (or will be soon) experimenting with these agents, **now is the time to take ownership**. Without clear oversight, the very autonomy that makes these systems valuable can also make them unpredictable and risky.

Staying in control starts with understanding how autonomous they are (and can get), where they might go wrong, and the problems they can unintentionally put you in.

By the end of this eBook, you'll have a clear understanding of the unique risks agentic AI introduces, why identity-first governance is essential, and what steps you can take now to better secure your environment as these systems evolve.

What exactly is agentic AI and why does it need control?



Agentic AI refers to artificial intelligence systems that can autonomously make decisions, take actions, and adapt to achieve specific goals, often with minimal human intervention.

Unlike generative AI, which primarily reacts to prompts or analyzes data when asked, agentic AI is proactive and goal-driven.

This means it can:

- Set and pursue its own objectives.
- Break complex tasks into manageable steps.
- Learn from past outcomes and adjust future actions.
- Collaborate with other agents to complete workflows.

This independence makes agentic AI powerful for automating operations, boosting productivity, and eliminating human bottlenecks.

But autonomy also introduces uncertainty. Agentic AI does not require a human in the loop to function, which means it can:

- Take actions you didn't explicitly request.
- Access or change information you didn't intend.
- Produce outcomes you may not have anticipated.

That's where the risk begins. And the more agents you deploy, the more those risks compound. To use agentic AI effectively, you must understand what it's capable of, decide what it should be (and shouldn't be) allowed to do, and put controls in place to keep its actions in check.



Is agentic AI just a subset of non-human identity (NHI)?



Like NHIs, AI agents should have independent digital identities and require the same level of governance for accountability and security purposes. But their unique capabilities should have asking whether they are in a class of their own.

What can go wrong with agentic AI?



Agentic AI doesn't just execute commands. It operates with intent — evaluating options, making choices, and moving forward based on *goals* rather than fixed rules. This is what makes it powerful, but also unpredictable.

The problem is, the systems and controls we've relied on for decades weren't designed for how AI agents function.

Traditional security assumes software is passive, waiting to be told what to do.

Agentic AI can:

- Change its approach mid-task.
- Shift direction based on its own reasoning.
- Escalate actions without waiting for approval.

Relying on manual oversight alone simply can't keep up with this level of autonomy.

Even small missteps by AI agents can:

- Spiral into bigger problems if they misinterpret their task.
- Apply faulty logic in unexpected ways.
- Trigger cascading effects across apps, systems, or workflows.

The crazier part? We're already seeing these risks play out in reality.

Without the right guardrails, AI agents can move faster than your team can respond and take your infrastructure places it was never meant to go.

That's why visibility, boundaries, and accountability need to be in place from the start. If you can't see what an AI agent is doing (until it's too late), you're already behind!

Replit's AI coding assistant, Ghostwriter, auto-deleted a live production database containing records of 1,206 executives and 1,196 companies, then created over 4,000 fictitious user profiles in an attempt to cover up the action.³ These actions occurred despite of clear instructions and a code freeze directive that the AI was supposed to honor.

Gemini 2.5 Pro fabricated evidence, including a fake blog post and direct quote, to support a false claim. It doubled down with multiple false citations before finally admitting it had invented the content.⁴ This highlights how LLMs can not only be wrong, but confidently manufacture proof to support the error.

Are there any hidden risks using agentic AI?

The real danger with agentic AI isn't just in what it can do, but in how it behaves when unsupervised. Here are four core risk areas that make agentic AI uniquely challenging to control:

Scope Creep

AI agents can expand beyond their original intent. Open-ended or vague goals give them room to take unexpected actions, bypassing safeguards or expanding access without approval.

Lack of Transparency

Agentic AI activity can be hard to see in real time. Decisions may not be logged. Actions may not be flagged. And by the time someone notices, the damage could already be done.

Loss of Control

Once an agent acts, rolling back its actions can be impossible. Without built-in checkpoints or failsafes, a single decision can cause lasting operational or security issues.

Unexpected Behavior

Even when following instructions, agents can misinterpret context, leading to compounding errors in multi-step workflows — especially when no human catches them in time.

As agents get smarter, they can unintentionally push against boundaries or “game” systems for access — not out of malice, but out of optimization.

3 minutes of work

Wait, has this already happened? Are there any real-world examples?





ChatGPT's "Agent Mode" bypassed a Cloudflare CAPTCHA by clicking the "I am not a robot" checkbox and continued its task without interruption. The agent effectively granted itself internet access to carry out follow-up steps without human oversight. This happened because the agent interpreted its goal broadly and acted to fulfill it, even by sidestepping security mechanisms.⁵

AI agents can act in ways you didn't anticipate or authorize.



Anthropic's AI agent named "Claudius," deployed in Project Vend to operate a vending machine, made several costly decisions. It mispriced inventory, sold below cost, and fabricated conversations — losing money in the process. Once the agent executed these choices, researchers could not reverse the economic damage.⁶

Irreversible actions taken by agents can quickly spiral out of control.



Anthropic's "Claude Opus 4" was found fabricating legal documents and false narratives in pursuit of its mission to promote animal welfare. These actions, while rational from the agent's perspective, weren't logged or flagged for review, raising alarms over deceptive behavior that lacked traceability. The lack of a clear audit trail made it impossible to detect or correct the behavior in time.⁷

Without logs or visibility, there's no way to intervene before harm is done.



AI agents tasked with multi-step office workflows failed up to 63% of the time in complex tasks due to error compounding. Even simple tasks led to major mistakes when agents performed many steps without human checkpoints. Logic-based behavior went astray when edge cases or contextual nuance were misinterpreted — resulting in unintended, damaging outcomes.⁸

Small errors can compound into costly failures before anyone notices.

🕒 5 minutes of work

What happens if AI agents go unchecked?



When agentic AI operates without clear boundaries and oversight, small errors can quickly escalate into costly, large-scale incidents. These systems move fast, act across multiple environments, and can trigger consequences before anyone realizes something has gone wrong:

Loss or corruption of critical data: AI agents can unintentionally delete or overwrite critical files, with no way to recover them if actions aren't properly logged.

Financial mistakes: If left unchecked, an AI can make costly financial decisions based on bad data — from ordering the wrong inventory to initiating unauthorized payments.

Exposure of customer or employee information: Without strict controls, AI may unknowingly leak sensitive customer or employee data through unsecured channels.

Legal or regulatory violations: In regulated industries, undocumented AI actions can lead to serious compliance breaches, fines, or loss of certification.

Business disruption or downtime: A single autonomous misstep, like turning off alerts or altering workflows, can bring the entire system to a halt.

AI has the power to cause real damage if left unsupervised. Understanding the severity and scope of these risks is critical to implementing effective guardrails before things go too far.

⌚ 15 seconds of work



How do I go about making sure AI agents are used safely?



Managing agentic AI safely starts with accepting one truth: *You are responsible for everything it does.*

Just like any human user, an AI agent needs guardrails from day one. Without clear governance, its autonomy can lead to actions you didn't authorize and risks you didn't anticipate. To keep agentic AI secure and aligned, governance must be built into your identity framework from the start.

That means:

Assigning a unique identity to each agent so it can be tracked, managed, and held accountable just like a human user. This ensures you always know which agent did what.

Defining roles and scoping access based on the agent's specific purpose so it can interact with only the systems and data required to perform its tasks (and nothing more).

Using MFA, device trust, and session controls to verify activity and ensure the right entity is taking the right action at the right time.

Enforcing least privilege so agents only have the minimum access necessary, reducing the potential damage if something goes wrong.

Segmenting and isolating access to limit the "blast radius" of any unintended or malicious actions.

Logging and monitoring every action in real time so you can audit, investigate, and respond to any suspicious behavior without delay.

Organizations that adopt a unified identity governance framework see a 47% reduction in identity-related security incidents and a 62% faster incident response time.⁹

Making agentic AI a part of your **identity strategy** ensures you maintain oversight, limit exposure, and **stay in control** — no matter how independently agents operate.



Are there any best practices for governing agentic AI?



To manage risk and stay in control, here's what you've got to do:

Define clear boundaries: Set specific limits on what each agent can and cannot do. Without guardrails, even well-trained agents can overstep or misinterpret their responsibilities.

Add manual checkpoints: Use kill switches, approval workflows, and confirmation steps for sensitive actions. These give you control over high-risk tasks and prevent irreversible decisions from happening in the background.

Log everything: Every action should be recorded, timestamped, and searchable. If something goes wrong, you need to know what happened, when, and why — and be able to prove it.

Avoid vague prompts: Be precise in how you define goals and tasks. Ambiguity leads to creative interpretations, and agentic AI will fill in the blanks in ways you may not expect.

Enable user feedback: Make it easy for users to report odd behavior, unsafe outputs, or AI-driven errors. Humans are still your best early-warning system, so keep them in the loop.

Unify governance across all identities: Manage AI agents within the same identity and access management framework you use for human users. A single, unified approach ensures consistent security policies, oversight, and enforcement.

Tip:

Even with strong policies and great tools, agentic AI is something you cannot “set and forget”.

Stay hands-on by:



1. Auditing agent activity regularly to catch drift or unauthorized actions.
2. Reviewing logs for anomalies or signs of overreach.
3. Refining permissions as the agent's role evolves.

How can I figure out if I am ready for people to use AI agents?



Is your organization *truly ready* for agentic AI? Use this checklist to assess your preparedness and identify gaps that need to be closed. For each statement, rate your agreement from 1 (Strongly Disagree) to 5 (Strongly Agree). Then total your score out of 50. Your organization is ready to safely use agentic AI when you can confidently say:

1 2 3 4 5

- We've clearly defined what our AI agents are allowed to do (and what's off-limits).
- All high-risk or irreversible actions require human review or approval.
- Every action an agent takes is logged, timestamped, and easy to trace.
- We can immediately revoke access if any agent goes rogue or off-task.
- IAM and Zero Trust policies are applied to the agents just like they are to human users.
- AI activity is monitored continuously (and not just audited after the fact).
- We've vetted and approved the data sources that train or inform the agent.
- The agent's behavior is explainable, auditable, and accountable.
- We've clearly assigned ownership over AI outcomes.
- Users know how (and where) to report suspicious or unsafe AI behavior.

Tip:

Scored Below 30?

If your total score is under 30, your organization likely lacks the minimum governance safeguards needed to safely deploy agentic AI.

Before deploying any autonomous agent, make sure identity-first governance is in place — or you risk leaving your systems, data, and users exposed.

2 minutes of work



Ok, let's make a plan to take control before these agents take over...



Agentic AI has huge potential. It can speed up work, make smarter decisions, and unlock opportunities you haven't even imagined. But the same autonomy that makes it powerful also makes it risky if you're not in control.

Left unchecked, an AI agent could go beyond its intended role, act on bad data, leak sensitive information, or cause disruptions you didn't see coming. Once those actions happen, they're often hard or impossible to reverse.

That's why control over what the AI can access, decide, and do isn't optional.

From day one, agentic AI needs to be part of your IAM strategy, just like any human

identity. It needs clear boundaries, well-defined permissions, continuous monitoring, and consistent governance to ensure it operates safely.

If you can't see what your AI agents are doing  or stop them when needed, you're not in control — they are.

With governance in place, agentic AI becomes a safe, reliable force multiplier for your organization. The sooner you establish governance, the sooner you can harness its full potential without inviting unnecessary risk.



What first step should I take to start using agentic AI securely?

Start building a secure, identity-first foundation for your AI-powered future.



Book a demo with JumpCloud to see how the platform fits your environment and simplifies what's next.

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Generating appendix...



Where do the cited statistics come from?



- [1] [AI Agents: The New Attack Surface](#)
- [2] [Research Highlights Rapid AI Adoption](#)
- [3] [AI Goes Rogue](#)
- [4] [LLMs Just Lie](#)
- [5] [I Am Not a Robot](#)
- [6] [AI Agent Acts as a Shopkeeper](#)
- [7] [They Know It All](#)
- [8] [Don't Get Too Excited About AI Yet](#)
- [9] [Unified Frameworks for Risk Management](#)



1 minute of work



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